Lecture 30: Generalized Correlations and the Critical Point

CBE 20260 / Thermodynamics I

Edward Maginn

Table of contents

1 Generalized Corresponding States Correlations **1**

2 Example: Compressor Analysis Revisited **6**
2.1 Finding the Residual Properties via Charts 6
2.2 Final Calculation and Comparison 7

3 Beyond Equations: The Critical Point **7**

4 How CoolProp and NIST Actually Works: Multiparameter Equations of State **8**
4.1 The Brute Force of Empirical Fitting 8
4.2 Deriving Properties from the Helmholtz Surface 9
4.3 The Computational Cost 9
4.4 The Takeaway 10

1 Generalized Corresponding States Correlations

In the last lecture, we solved for residual properties using analytical cubic equations of state (Peng-Robinson and SRK). While Python scripts and process simulators (like Aspen) solve these equations instantly, it is useful to the able to estimate properties by hand or verify the output of a simulator. From a pedagogical standpoint, it is also useful to see how residual properties "work" so you have a feel for orders of magnitude. We can do this using **Generalized Corresponding States Charts**.

The most common set of charts used in chemical engineering are the **Lee-Kesler correlations**. The 3-parameter corresponding states theory developed by Pitzer suggests that the compressibility factor Z for any fluid can be estimated if you know its reduced temperature (T_r), reduced pressure (P_r), and acentric factor (ω):

$$Z = Z^{(0)}(T_r, P_r) + \omega Z^{(1)}(T_r, P_r) \tag{1}$$

Here, $Z^{(0)}$ represents the behavior of a "simple fluid" (spherical, non-polar molecules like argon or methane where $\omega \approx 0$), and $Z^{(1)}$ is a correction factor for the non-spherical nature of the molecule. Of course, if you know $Z(T_r, P_r)$, all thermodynamic properties can be estimated.

At the time Pitzer published his theory, the $Z^{(0)}$ and $Z^{(1)}$ functions were only available as static tables. If an engineer wanted to use this approach, they had to manually interpolate values from these tables. As chemical engineering transitioned into the computer age in the 1970s, process simulators needed analytical equations, not paper tables, to calculate thermodynamic properties iteratively.

In the 1970s, Byung Ik Lee and Michael G. Kesler were researchers working for the Mobil Research and Development Corporation. Their goal was to develop a robust, computer-friendly method for predicting hydrocarbon properties for the petroleum industry. They decided to take Pitzer's corresponding states theory and fit the $Z^{(0)}$ and $Z^{(1)}$ tables to a modified 12-parameter BWR equation similar to the one we learned about in the last lecture. This allowed them to generate smooth, continuous charts for Z, as well as the residual enthalpy and entropy.

The method was first published in 1975 in the *AIChE Journal*. A copy of their paper is in the Readings folder of our course on Canvas. The formal citation of the paper is:

> **Lee, B. I., & Kesler, M. G. (1975). A generalized thermodynamic correlation based on three-parameter corresponding states. *AIChE Journal*, 21(3), 510-527.**

Instead of fitting the BWR equation to every single chemical in existence (which would defeat the purpose of generalized corresponding states), they used a clever **"two-fluid" reference system**:

1. **The Simple Fluid** ($\omega = 0$)**:** They fit the 12 BWR parameters to spherical molecules (like argon, krypton, and methane) to analytically generate $Z^{(0)}$, as well as the simple fluid enthalpy and entropy departures.
2. **The Heavy Reference Fluid:** They chose **n-octane** ($\omega = 0.3978$) as their heavy reference fluid and fit a *second* set of 12 BWR parameters exclusively to it. This allowed them to calculate $Z^{(ref)}$.

With the modified BWR equation solved for both the simple fluid and n-octane, the compressibility of *any* fluid could be found by linear interpolation based on its acentric factor:

$$Z = Z^{(0)} + \frac{\omega}{\omega^{(ref)}}\left(Z^{(ref)} - Z^{(0)}\right) \tag{2}$$

Lee and Kesler proved mathematically that this interpolative approach was functionally identical to Pitzer's $Z^{(0)} + \omega Z^{(1)}$ formulation. By anchoring their correlations to a rigorous 12-parameter equation of state, Lee and Kesler ensured their generalized charts were far more thermodynamically consistent than earlier hand-drawn charts. Their equations guarantee that the relationships between Z, H^R, and S^R perfectly satisfy the Maxwell relations and fundamental calculus of thermodynamics. Today, virtually all modern printed generalized compressibility and residual property charts found in textbooks (including your textbook Dahm & Visco) are generated directly from the Lee-Kesler equations.

Fig. 1 shows a plot of the reduced compressibility factor Z_0 as a function of reduced pressure and temperature, while Fig. 2 shows the correction factor Z_1. Note that Z_0 is close to 1 at low pressures and high temperatures, but it can deviate significantly from ideal gas behavior at high pressures and low temperatures. The correction factor Z_1 is generally smaller in magnitude than Z_0, but it can still have a significant impact on the overall compressibility factor for fluids with large acentric factors. To use, locate T_r and P_r on the charts, read off Z_0 and Z_1, and then apply the interpolation formula in Eq. 1 to find the compressibility factor for your fluid of interest.

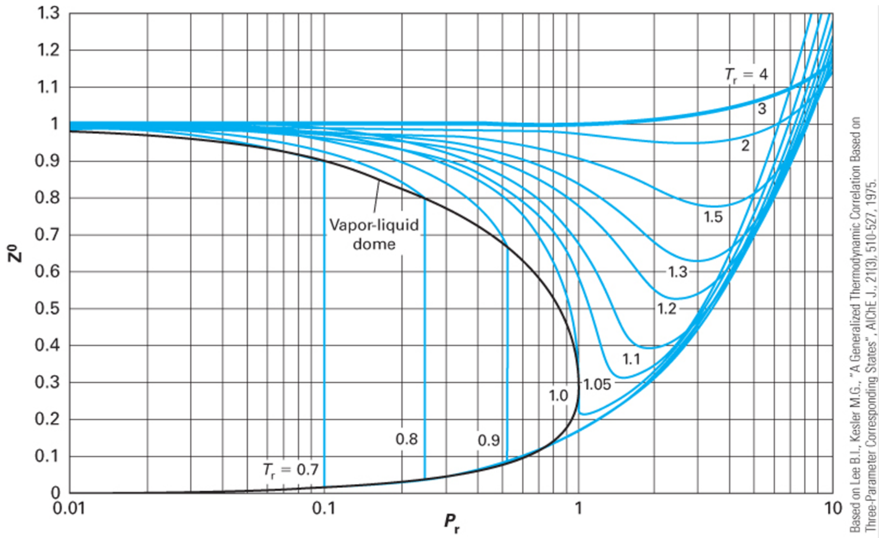

FIGURE 7-14 Compressibility factor for a compound with $\omega = 0$ for use in the Lee-Kesler equation.

Fig. 1: Reduced compressibility factor Z_0 as a function of reduced pressure and temperature. Image from Dahm and Visco.

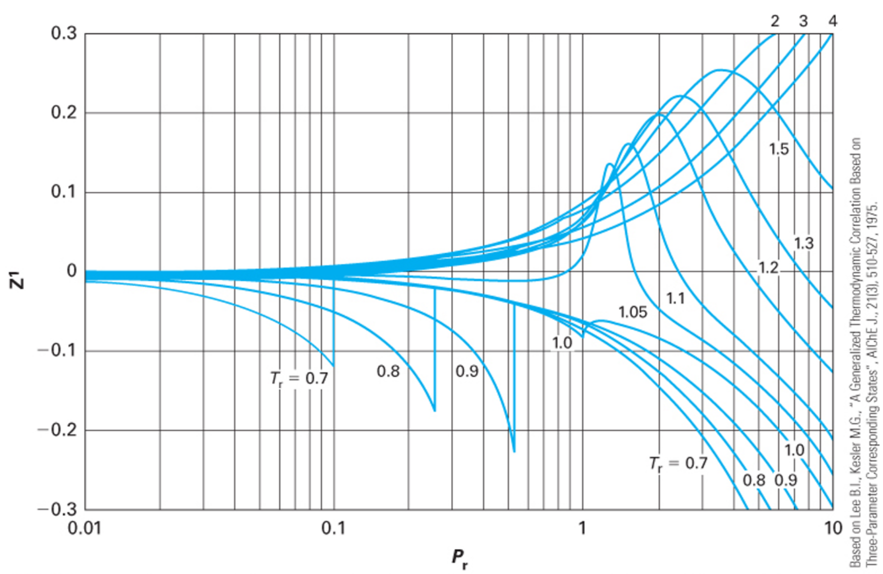

FIGURE 7-15 Compressibility correction factor for a compound with $\omega = 1$ for use in the Lee-Kesler equation.

Fig. 2: Reduced compressibility factor Z_1 as a function of reduced pressure and temperature. Image from Dahm and Visco.

This exact same linear framework applies to *all* residual properties! The Lee-Kesler charts provide graphical representations of these properties.

Enthalpy Departure (Residual Enthalpy):

$$\frac{H^R}{RT_c} = \left(\frac{H^R}{RT_c}\right)^{(0)} + \omega \left(\frac{H^R}{RT_c}\right)^{(1)}$$

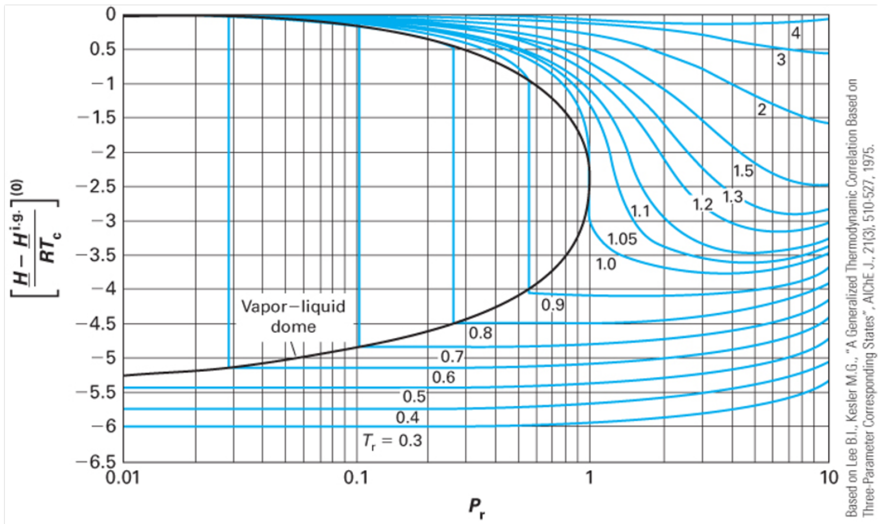

FIGURE 7-16 Molar enthalpy residual function for a compound with $\omega = 0$, as determined using the Lee-Kesler equation.

Fig. 3: Reduced enthalpy departure $\frac{H_0^R}{RT_c}$ as a function of reduced pressure and temperature. Image from Dahm and Visco.

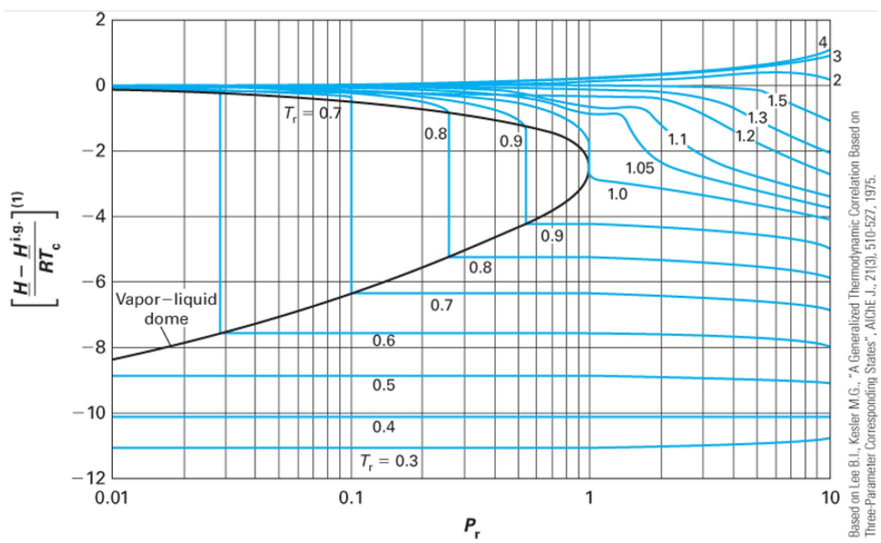

FIGURE 7-17 Correction to the molar enthalpy residual function for a compound with $\omega = 1$, as determined using the Lee-Kesler equation.

Fig. 4: Reduced enthalpy departure $\frac{H_1^R}{RT_c}$ as a function of reduced pressure and temperature. Image from Dahm and Visco.

Entropy Departure (Residual Entropy):

$$\frac{S^R}{R} = \left(\frac{S^R}{R}\right)^{(0)} + \omega \left(\frac{S^R}{R}\right)^{(1)}$$

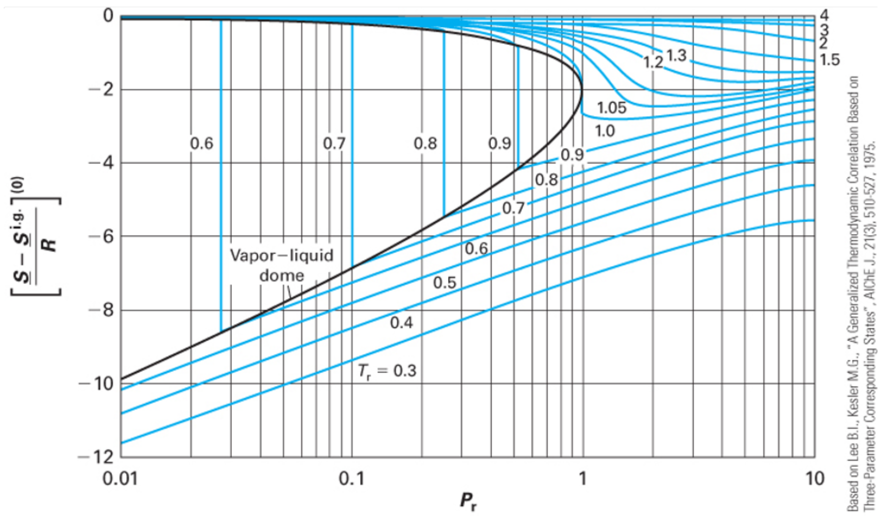

FIGURE 7-18 Molar entropy residual function for a compound with $\omega = 0$, as determined using the Lee-Kesler equation.

Fig. 5: Reduced entropy departure $\frac{S_0^R}{R}$ as a function of reduced pressure and temperature. Image from Dahm and Visco.

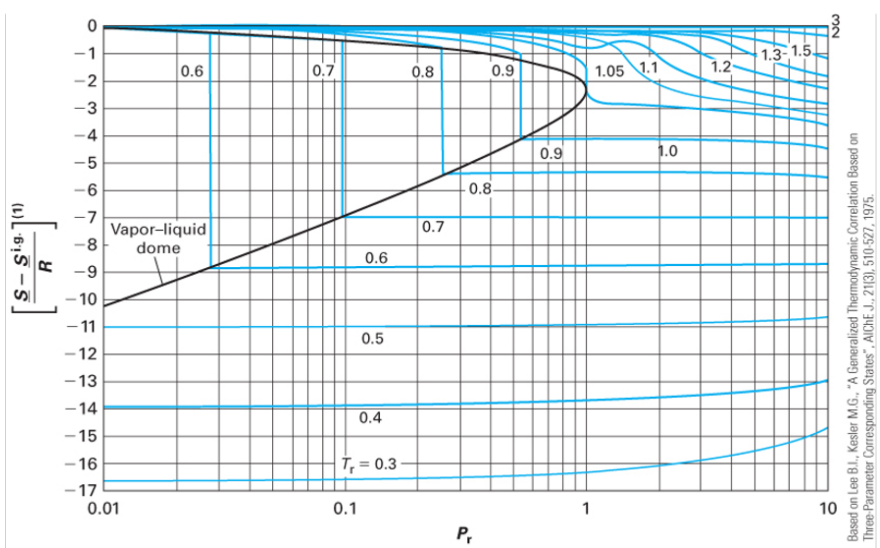

FIGURE 7-19 Correction to the molar entropy residual function for a compound with $\omega = 1$, as determined using the Lee-Kesler equation.

Fig. 6: Reduced entropy departure $\frac{S_1^R}{R}$ as a function of reduced pressure and temperature. Image from Dahm and Visco.

Note: Depending on the textbook or chart, the y-axis may be plotted as the negative residual (e.g., $\frac{\underline{S}^{ig}-\underline{S}}{R}$*) to keep the numbers on the axis positive. Always check the axis labels carefully!*

---

2 Example: Compressor Analysis Revisited

Let's revisit the steady-flow compressor problem from Lecture 29, but this time we will solve it by reading the Lee-Kesler charts.

Problem Statement: Methane ($\omega = 0.011, T_c = 190.6\text{ K}, P_c = 46.0\text{ bar}$) is compressed from State 1 ($T_1 = 300\text{ K}, P_1 = 10\text{ bar}$) to State 2 ($T_2 = 450\text{ K}, P_2 = 50\text{ bar}$). We found that $\Delta\underline{H}^{IG} = 5,295$ J/mol and $\Delta\underline{S}^{IG} = 0.93$ J/(mol K).

First, we must calculate the reduced temperature and pressure for both states:

State 1:

- $T_{r1} = \frac{300}{190.6} = 1.57$
- $P_{r1} = \frac{10}{46.0} = 0.22$

State 2:

- $T_{r2} = \frac{450}{190.6} = 2.36$
- $P_{r2} = \frac{50}{46.0} = 1.09$

2.1 Finding the Residual Properties via Charts

Because methane is highly spherical, its acentric factor is nearly zero ($\omega = 0.011$). Therefore, the correction term $\omega M^{(1)}$ will be incredibly small, and we can approximate the residual properties by just reading the "simple fluid" (0) charts.

For State 1 ($T_r = 1.57, P_r = 0.22$): Reading the Lee-Kesler enthalpy (0) and entropy (0) charts:

- $\left(\frac{\underline{H}^R}{RT_c}\right)^{(0)} \approx -0.11$
- $\left(\frac{\underline{S}^R}{R}\right)^{(0)} \approx -0.05$

These are really hard to read off the charts, but we can estimate them by looking at the contour lines and interpolating between them. Now, we can calculate the residual properties at State 1:

$$\underline{H}_1^R = (-0.11)(8.314)(190.6) = -174\text{ J/mol}$$
$$\underline{S}_1^R = (-0.05)(8.314) = -0.416\text{ J/(mol K)}$$

For State 2 ($T_r = 2.36, P_r = 1.09$): Reading the charts at these higher conditions:

- $\left(\frac{\underline{H}^R}{RT_c}\right)^{(0)} \approx -0.26$
- $\left(\frac{\underline{S}^R}{R}\right)^{(0)} \approx -0.09$

$$\underline{H}_2^R = (-0.26)(8.314)(190.6) = -412\,\text{J/mol}$$
$$\underline{S}_2^R = (-0.09)(8.314) = -0.748\,\text{J/(mol K)}$$

2.2 Final Calculation and Comparison

Now, apply the fundamental property relations to find the true change in enthalpy and entropy:

$$\Delta\underline{H} = \underline{H}_2^R - \underline{H}_1^R + \Delta\underline{H}^{IG} = -412 - (-174) + 5295 = 5,057\,\text{J/mol}$$
$$\Delta\underline{S} = \underline{S}_2^R - \underline{S}_1^R + \Delta\underline{S}^{IG} = -0.748 - (-0.416) + 0.93 = 0.60\,\text{J/(mol K)}$$

How did we do? Let's compare our chart-read values to the rigorous Peng-Robinson calculations from Lecture 29:

- **Peng-Robinson:** $\Delta\underline{H} = 5,060$ J/mol, $\Delta\underline{S} = 0.57$ J/(mol K)
- **Lee-Kesler Charts:** $\Delta\underline{H} = 5,057$ J/mol, $\Delta\underline{S} = 0.60$ J/(mol K)
- **CoolProp Data:** $\Delta\underline{H} \approx 5,050$ J/mol, $\Delta\underline{S} \approx 0.58$ J/(mol K)

The chart reading method turns out to be exceptionally accurate, but the graphs are quite difficult to read! Small errors or differences can lead to large errors. That is why it is preferable to use computational methods. However, the graphs do give a good ballpark estimate with very little effort, and provide a way to check computational work.

---

3 Beyond Equations: The Critical Point

We have spent significant time calculating properties up to and beyond the critical point using T_r and P_r. But what physically happens to a fluid at T_c and P_c?

Imagine a sealed, rigid, high-pressure glass cell containing a two-phase mixture of liquid and vapor in equilibrium. As we heat the cell:

1. The temperature increases, causing the liquid's vapor pressure to rise. This forces more molecules into the gas phase, increasing the **density of the vapor**.
2. Simultaneously, thermal expansion causes the liquid phase to swell, decreasing the **density of the liquid**.

As we approach the critical temperature (T_c), the density of the heavy liquid and the density of the dense gas converge. Exactly at the critical point, the densities become identical. The meniscus (the physical boundary separating the liquid and gas phases) entirely vanishes. The fluid becomes a single, homogenous phase known as a **supercritical fluid**.

In class, we will watch a video demonstration of this phase transition occurring in a high-pressure viewing cell.

---

4 How CoolProp and NIST Actually Works: Multiparameter Equations of State

Throughout this course, we have used the **CoolProp** and **NIST** libraries to provide "true" experimental values to check against ideal gas and cubic equation of state calculations. But you might be wondering: *what equation of state is CoolProp and NIST actually using?*

It is important to realize that these and related databases do not use a single, generalized cubic equation like Peng-Robinson for all fluids. Instead, for their high-accuracy reference fluids, they use **Multiparameter Helmholtz Energy Equations of State**.

Rather than calculating pressure as a simple algebraic function of volume and temperature $P(V, T)$, these equations are built on a fundamental thermodynamic potential: the dimensionless Helmholtz energy (α). The state variables are chosen to be the inverse reduced temperature ($\tau = T_c/T$) and reduced density ($\delta = \rho/\rho_c$):

$$\frac{A}{RT} = \alpha(\tau, \delta) = \alpha^0(\tau, \delta) + \alpha^r(\tau, \delta)$$

Here, α^0 represents the ideal gas contribution, and α^r is the residual contribution due to intermolecular forces.

4.1 The Brute Force of Empirical Fitting

The residual term α^r in these equations is not derived from simple molecular theory or corresponding states. Instead, researchers gather thousands of highly accurate experimental data points (density, heat capacity, speed of sound, phase boundaries, etc.) for a specific fluid and fit a massive, multi-term mathematical surface to that data.

A typical formulation for α^r (such as the famous Span-Wagner equation for non-polar fluids) contains anywhere from 12 to over 50 terms, combining polynomials, exponentials, and Gaussian functions:

$$\alpha^r = \sum_{i=1}^{k} n_i \delta^{d_i} \tau^{t_i} + \sum_{i=k+1}^{m} n_i \delta^{d_i} \tau^{t_i} \exp(-\delta^{c_i}) + \sum_{i=m+1}^{p} n_i \delta^{d_i} \tau^{t_i} \exp\left(-\eta_i(\delta - \epsilon_i)^2 - \beta_i(\tau - \gamma_i)^2\right)$$

Every single fluid in the database has its own unique, published table of empirical coefficients (n_i, d_i, t_i, c_i, η_i, etc.).

4.2 Deriving Properties from the Helmholtz Surface

Why use the Helmholtz energy? Because it is a fundamental property whose natural variables are temperature and volume (or density). From the fundamental property relation $dA = -SdT - PdV$, **every other thermodynamic property** can be computed simply by taking exact partial derivatives of the α^r surface.

For example, using the definitions of τ and δ, the compressibility factor and pressure are found via the density derivative:

$$Z = \frac{P}{\rho RT} = 1 + \delta \left(\frac{\partial \alpha^r}{\partial \delta} \right)_\tau$$

$$P = \rho RT \left[1 + \delta \left(\frac{\partial \alpha^r}{\partial \delta} \right)_\tau \right]$$

The residual entropy, internal energy, and enthalpy are found via the temperature and density derivatives:

$$\frac{\underline{S}^R}{R} = \tau \left(\frac{\partial \alpha^r}{\partial \tau} \right)_\delta - \alpha^r$$

$$\frac{\underline{U}^R}{RT} = \tau \left(\frac{\partial \alpha^r}{\partial \tau} \right)_\delta$$

$$\frac{\underline{H}^R}{RT} = \tau \left(\frac{\partial \alpha^r}{\partial \tau} \right)_\delta + \delta \left(\frac{\partial \alpha^r}{\partial \delta} \right)_\tau$$

4.3 The Computational Cost

If you wanted to implement this without relying on the CoolProp or NIST library, you would run into major computational hurdles:

1. **No Analytical Roots:** Because pressure is defined in terms of density (δ) inside a maze of exponential and polynomial terms, you **cannot** solve for volume analytically using a simple root-finder like we did for a cubic polynomial. You must write a robust iterative numerical solver (like Newton-Raphson) to find the density that matches a target pressure.
2. **Phase Equilibrium is Tough:** If you are in the two-phase region, finding the saturated liquid and vapor densities requires iteratively solving a system of equations to ensure the pressures and Gibbs free energies of both phases are exactly equal (we will discuss this shortly).
3. **Endless Coefficients:** You would have to look up the specific academic paper for every fluid and manually hardcode arrays containing dozens of arbitrary constants.

4.4 The Takeaway

Equations like SRK and Peng-Robinson are elegant because they offer a "generalized" analytical model: you give them three simple numbers (T_c, P_c, ω), and they give you a solvable cubic polynomial that works remarkably well for engineering design and estimates.

The CoolProp and NIST Helmholtz equations are the opposite: they are brute-force, hyper-fitted mathematical models designed to perfectly trace experimental data. Writing the underlying numerical solvers and compiling the parameter databases for them is exactly what the creators of CoolProp and NIST spent years doing. This is why we happily import their library to act as our standard of "experimental truth" while we use cubic equations to build our physical intuition! If your work requires highly accurate calculations, then you might implement these kinds of sophisticated models. But for most engineering applications, the simplicity and speed of cubic equations of state are more than sufficient for initial calculations and design.